Final Project: Diabetes Progression Prediction (Group 4)

1. Detailed Analysis of Project Objectives and Problem Statement

The objective of the project is to use the data set to train and evaluate the k-Nearest Neighbors Regression and simple linear regression models. In order to predict the Glucose levels in patients.

While in the project description it suggests to predict “disease progression,” this project uses Glucose as the target variable instead of diabetes as the diabetes in the dataset is a classification in the dataset and is not suitable for regression algorithms, and glucose level is highly correlated with diabetes. The goal was to determine if a patient’s features like age, BMI, and blood pressure can accurately predict a current glucose level.

Critical Decisions:

- **Target Variable:** The code explicitly extracts the "Glucose" column as the target variable (y).
- **Zero Handling:** The rows with a target value of 0 were treated as invalid data and not used in the training. While the feature columns like age, BMI, and blood pressure containing missing data(0) were imputed using the mean of the non-zero values for that feature. This makes sure the model is trained on valid data.

2. Detailed Description of Data Handling and Modelling Components

2.1 Dataset Description

The analysis utilizes a "Diabetes data set". The raw data is loaded from diabetes.csv.

- **Target:** Glucose.
- **Features Used for kNN:** Pregnancies, Blood Pressure (BP), Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age.
- **Feature Used for Linear Regression:** Body Mass Index (BMI).

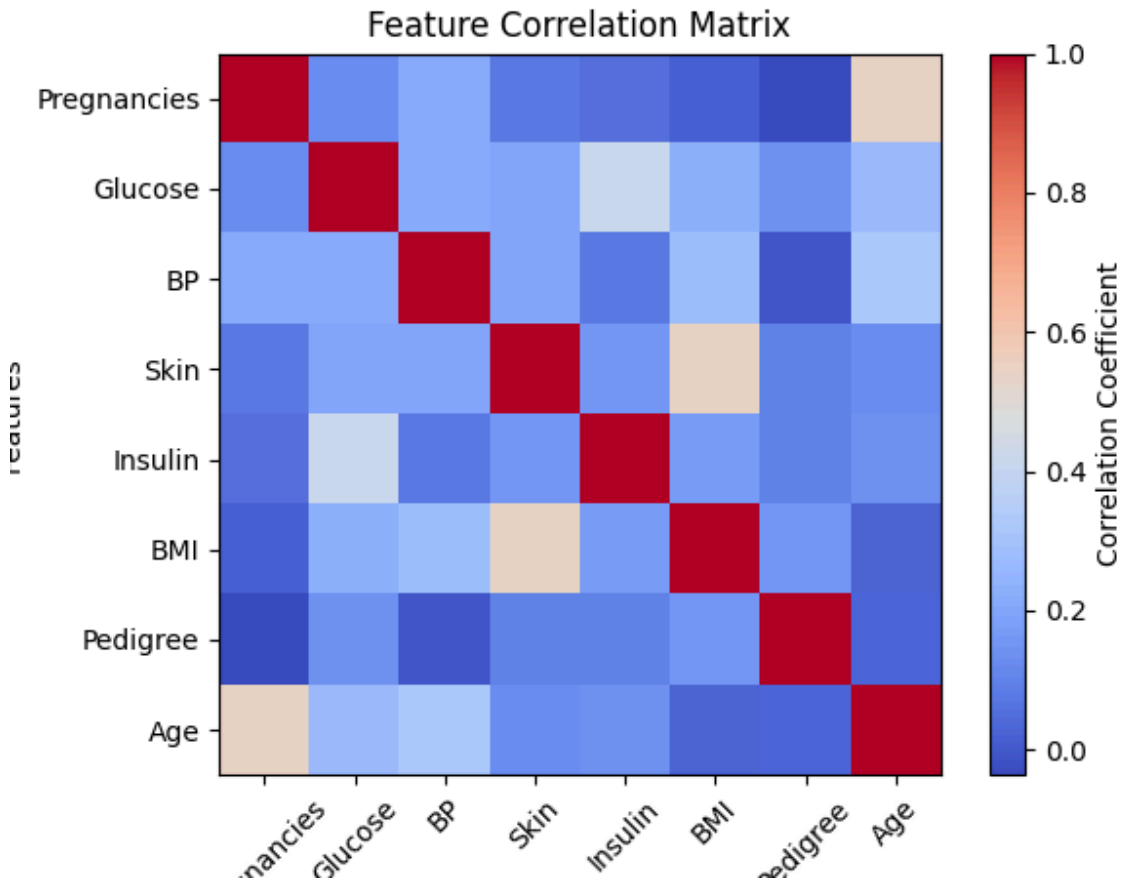
2.2 Pre-processing

1. **Data Cleaning:** The script goes through the CSV rows and discards any row when the target (Glucose) is invalid(0).
2. **Imputation:** For the features BP, Skin, Insulin, and BMI, values were assessed to contain missing data. To fill in the data a function(impute_zeros_with_mean) was written to replace the zeros with the average of all the valid entries in the category.

3. **Feature Transformation (kNN):** For the kNN model, all features were used. In order to make sure the weighting of the features was equal during the distance calculation the data was rescaled. The rescale function(rescale) uses Z-score normalization so that all feature data has a mean of 0 and a standard deviation of 1.
4. **Feature Selection (Linear Regression):** For Simple Linear Regression, the model used an isolated single feature. BMI was used as the independent variable to predict Glucose. Which tested the relationship between BMI and high blood sugar.

2.4 Feature Correlation Analysis

A correlation matrix was made to help understand the relationship between different variables and the target using the correlation_matrix function from the scratch.working_with_data module.



Correlation Matrix Heatmap of Diabetes Features

The Matrix Heatmap shows the linear relationships between all the variables in the cleaned dataset.

Analysis of Model Relevance:

- **kNN:** The matrix shows that the feature that is most correlated with Glucose is Insulin with a correlation coefficient of 0.58. This a strong positive relationship, and the rest of the features are moderate correlations making the full feature set a strong way to capture the best prediction.
- **kNN Advantage:** The correlation values show that the kNN has a bigger advantage by capturing the high correlation coefficient of insulin, suggesting it should be a better fit than the SLR model with BMI.

2.3 Model Setup

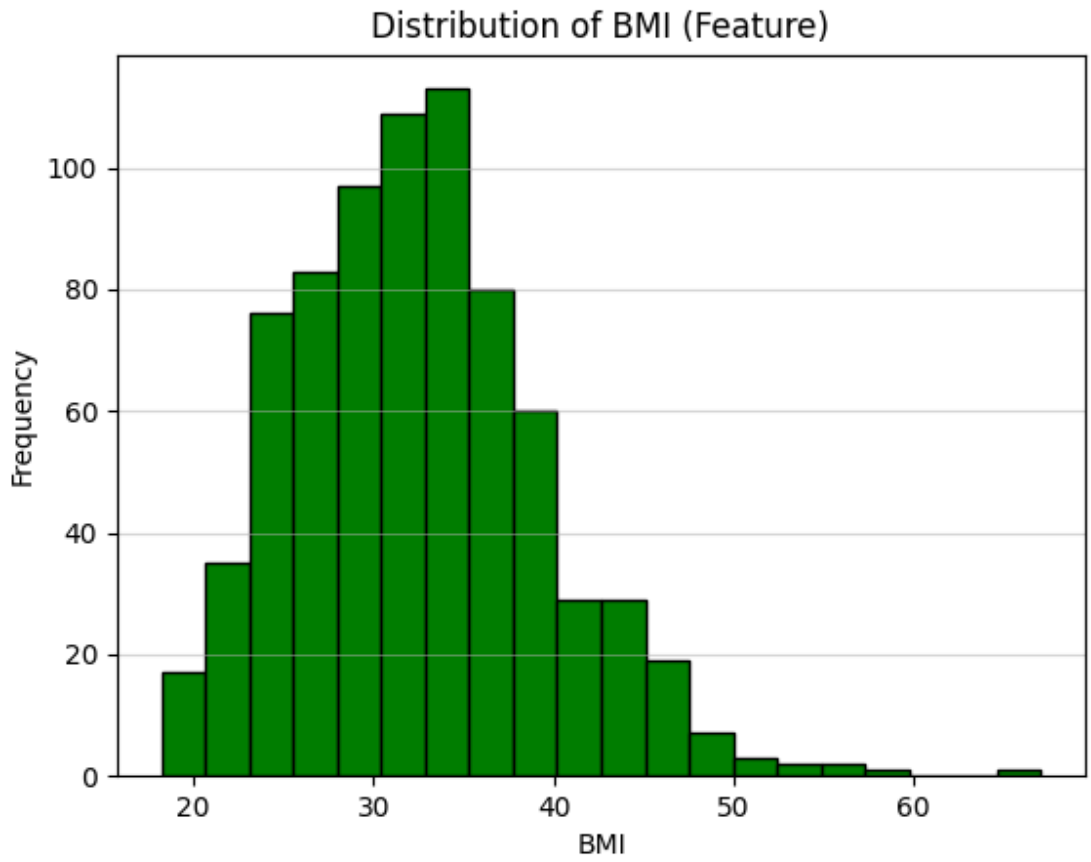
- **Data Splitting:** For both models, the data was split into 80% for a training set and 20% for a testing set to keep consistency when comparing the models.
- **kNN Regression:** The model predicts the glucose level of a new patient by calculating the Euclidean distance of all the training points. Then selecting the kNN and calculating the mean of their glucose levels. The code also tests for k values ranging from 1-30 using the elbow method to get the best k value.
- **Simple Linear Regression:** The model calculates the slope (beta) and intercept (alpha) using the least squares fit method on the training data.

3. Exploratory Data Analysis and Results

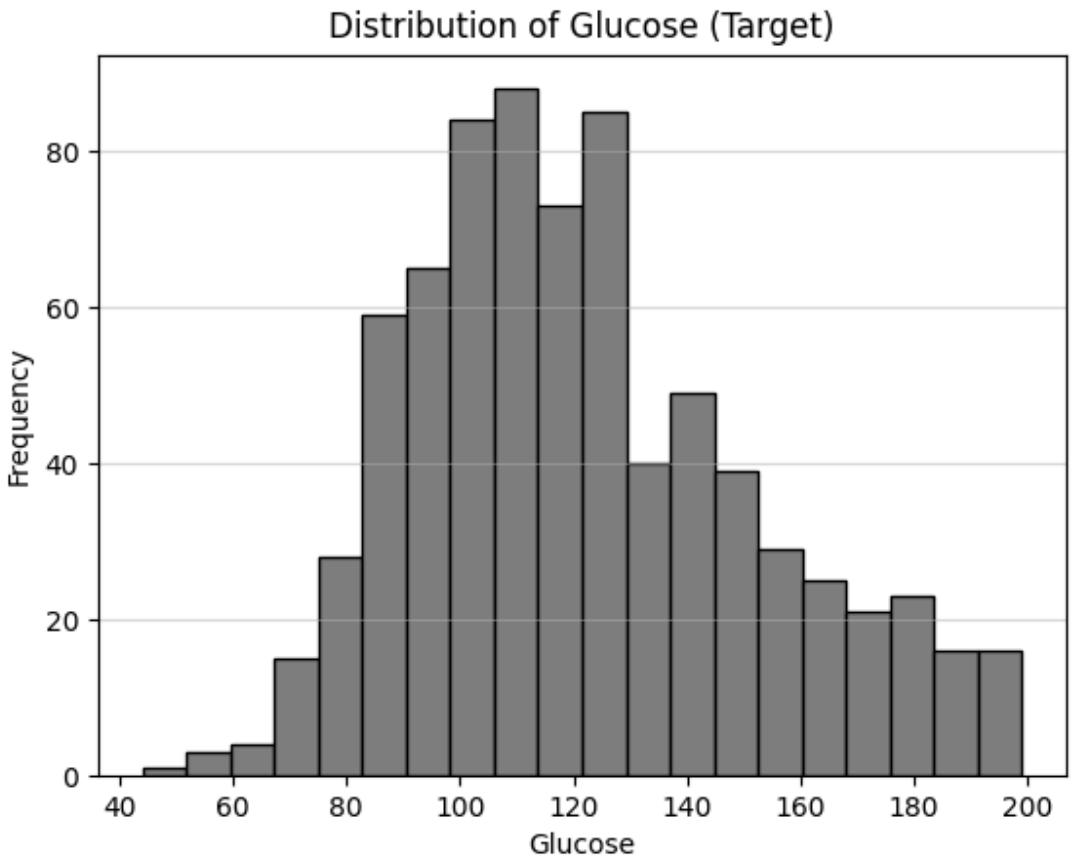
3.1 Exploratory Data Analysis

EDA was used to help understand the distribution of the BMI and Glucose variables used in the Simple Linear Regression model.

- **Distribution of BMI:** The histogram below shows the frequency distribution of Body Mass Index values in the dataset after imputation.



- **Distribution of Glucose:** The histogram below shows the spread of Glucose levels.



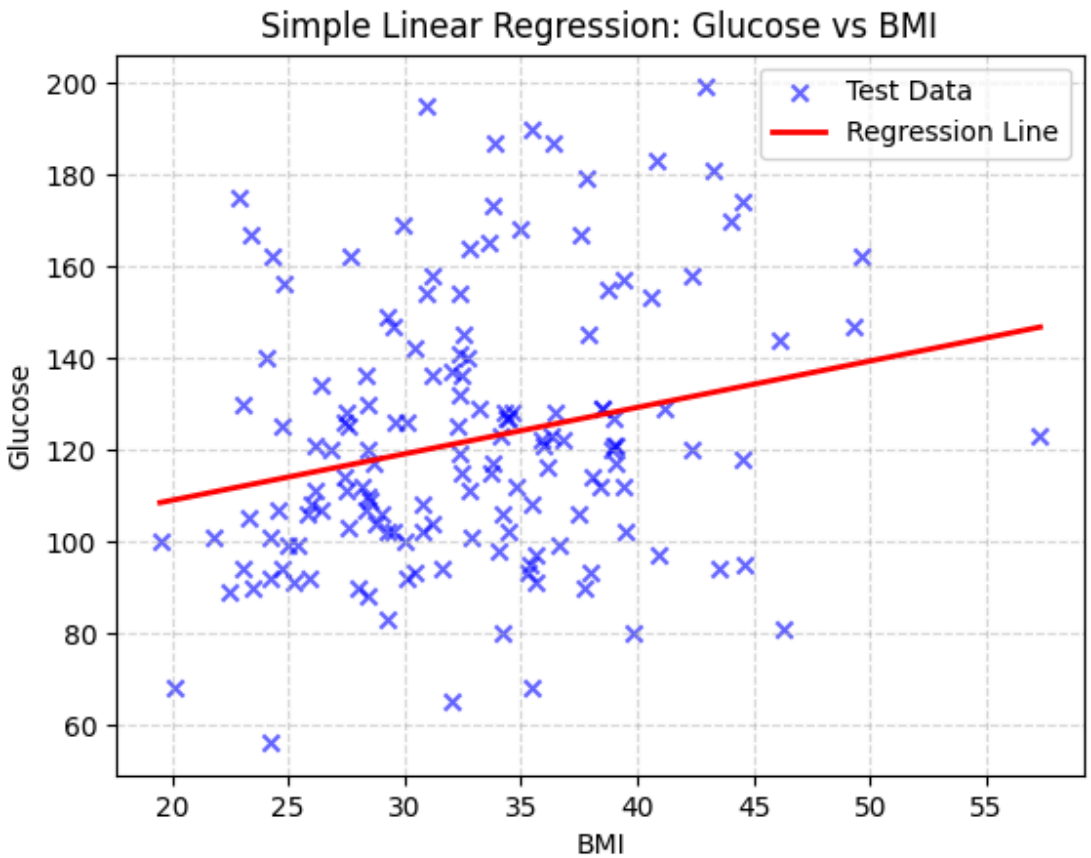
3.2 Simple Linear Regression Results

The SLR model attempts to predict Glucose based solely on BMI.

- Model Equation: Based on the training data, the linear relationship was calculated as: Glucose = 1.01(beta) * BMI + 88.69(alpha)

  This suggests that for every unit of BMI increased that Glucose levels increase by approximately 1.01.

- **Visualization:** The scatter plot below shows the testing data with the regression line.



Evaluation:

The model performance on the test set:

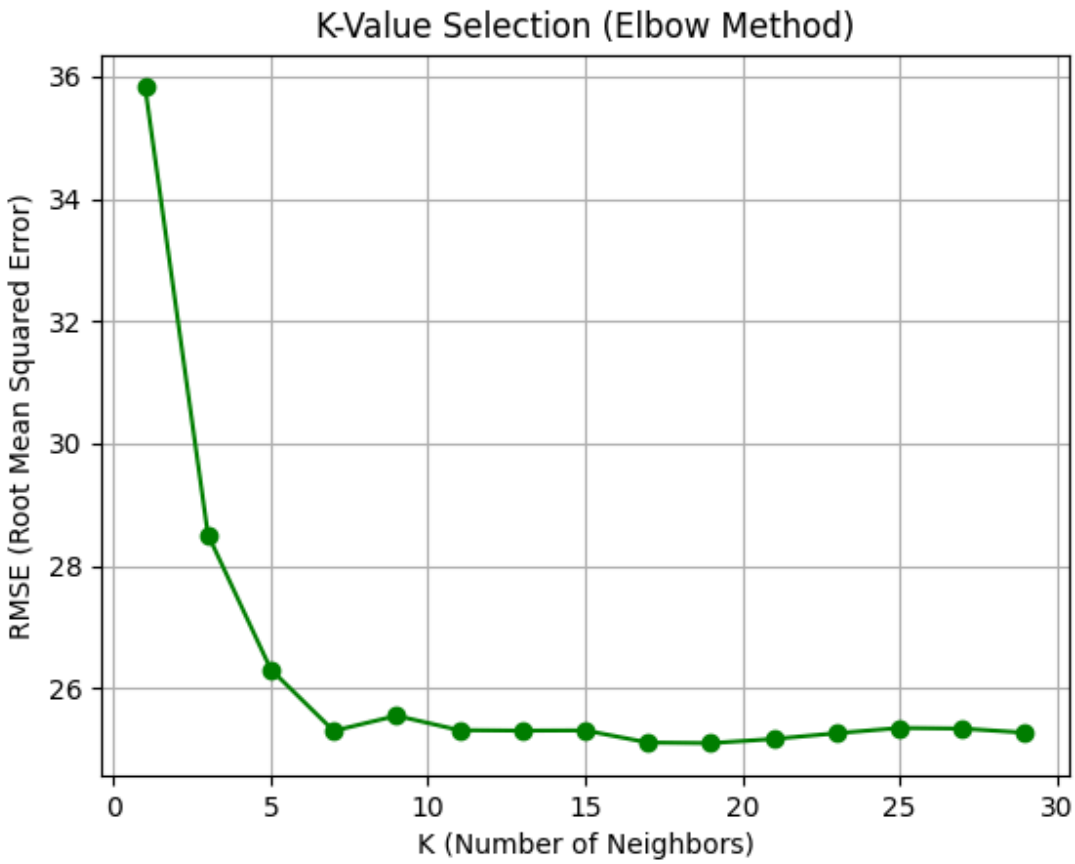
- **RMSE:** 28.00
- **R2 Score:** 0.06

Interpretation: The low R2 of 0.06 suggests that solely the BMI only accounts for about 6% of variance in Glucose levels. While it is positive, the BMI is not a strong predictor on its own.

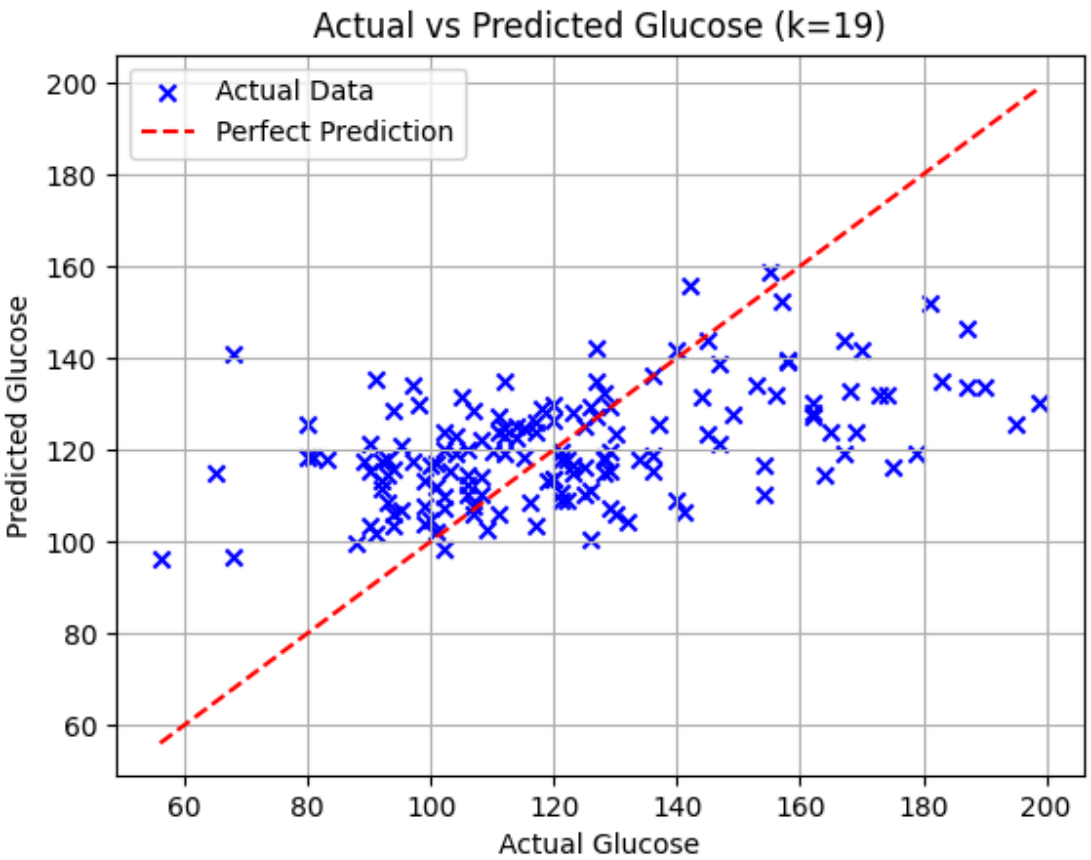
3.3 k-Nearest Neighbors (kNN) Results

The kNN model utilized all available features (Pregnancies, BP, Skin, Insulin, BMI, Pedigree, Age) to predict Glucose.

- **Hyperparameter Tuning (Elbow Method):** The model goes through k values from 1-31. The graph below then plots the RMSE with each k value to identify the best k number of neighbors.



- **Model Performance:** The model performed best with a k value of 19.
    - **Best RMSE:** 25.11
    - **Best R2$ Score:** 0.24
- **Visualization:** The scatter plot below compares the Actual Glucose values with the Predicted Glucose values that were generated by k=19



4. Discussion and Conclusions

4.1 Comparison of Models

This project evaluated two different ML models with different approaches. The SLR model using a single feature parametric model and the kNN using a multi-feature model.

- **Linear Regression:**
  - Performance: Poor (R2 = 0.06, RMSE = 28.00).
  - Pros: Clearly shows the relationship between BMI and Glucose making it easy to interpret.
  - Cons: Fails to capture how complex and factors of the disease.
- **kNN Regression:**
  - Performance: Significantly better than SLR(R2 = 0.24, RMSE = 25.11).
  - Pros: Uses multiple different features, which lets the model capture nonlinear patterns
  - Cons: Is considerably more computationally intense and harder to interpret what's happening behind the model

4.2 Interpretation and Implications

While the analysis confirms that BMI is correlated with Glucose, it is not a strong indicator when it is used alone. The kNN is a much better predictor due to using multiple features which improves the accuracy with a much higher variance of 24% compared to the 6% that the SLR model had.

In a clinical setting relying on a single metric to determine diabetes progression is largely insufficient. Though using multiple features like kNN regression is much more accurate. Even though a multiple features model is better with kNN a R2 of 0.24 suggests that there are many other factors that are not present in the dataset that play a role in predicting Glucose levels in patients.